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RMS Modeling and Control of a Grid-Forming E-STATCOM for Power System Stability in Isolated Grids

José Luis Rodríguez-Amenedo , Miguel E. Montilla-DJesus , Santiago Arnaltes *  and Francisco Arredondo 

Electrical Engineering Department, Universidad Carlos III de Madrid, 28005 Madrid, Spain; amenedo@ing.uc3m.es (J.L.R.-A.); mmontill@ing.uc3m.es (M.E.M.-D.); farredon@ing.uc3m.es (F.A.)

* Correspondence: arnalte@ing.uc3m.es

Abstract: This paper presents a comprehensive RMS-based phasorial model of an E-STATCOM with grid-forming (GFM) control, designed to improve power system stability in isolated grids. Unlike previous approaches, this model integrates a governor with an internal power system stabilizer (PSS) and an active current limiter (ACL) to enhance frequency regulation and mitigate oscillations. Additionally, an exciter with a nonlinear modulation function is introduced to optimize voltage regulation and reactive power support. A detailed conventional supercapacitor (SC) model is also incorporated, enabling dynamic DC-voltage control based on active power variations, improving frequency stability. The proposed E-STATCOM RMS model includes algebraic equations, dynamic governor and exciter models, supercapacitor-based energy storage control, and an advanced current-limiting strategy. Simulations are conducted on the Fuerteventura–Lanzarote (Canary Islands, Spain) power system, comparing the E-STATCOM with a synchronous condenser (SynCon) in frequency response, voltage regulation, and fault performance. The results show that the E-STATCOM improves frequency stabilization and energy efficiency while complying with grid codes. This study introduces a novel RMS-based modeling approach for GFM E-STATCOMs, bridging the gap between theoretical phasorial analysis and real-world applications. The findings confirm that E-STATCOMs are a viable alternative to SynCons, enhancing grid stability in high-renewable-penetration systems.



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Keywords: grid-forming control; E-STATCOM; DC-AC power conversion; supercapacitor-based energy storage; power system stability; frequency response; voltage regulation

1. Introduction

The increasing penetration of renewable energy sources (RESs) in power systems, particularly in weak grids, has introduced new stability challenges. The transition from conventional synchronous generators (SGs) to inverter-based resources (IBRs) such as wind farms, solar PV, energy storage systems (ESSs), and hybrid plants is reshaping the electric grid at an accelerated pace [1].

In conventional power systems, SGs and their auxiliary control elements provide essential grid services, including inertial response, which slows down the initial rate of change of frequency (RoCoF) after generation trips; fast frequency response (FFR) for frequency stabilization; voltage regulation; power oscillation damping; and negative sequence current injection [2]. However, as IBRs replace SGs, power systems are experiencing higher RoCoF and lower short-circuit current contributions, potentially compromising system strength and stability [3].

To address the stability challenges posed by the increasing penetration of IBRs, synchronous condensers (SynCons) have traditionally been deployed as a well-established

solution for grid strength enhancement. These rotating machines help stabilize voltage and provide inertial response, making them effective in weak grids. However, their high capital cost and long deployment time limit their feasibility for widespread adoption [4]. As an alternative, power electronics-based solutions such as E-STATCOMs have gained attention [5–7]. Unlike conventional STATCOMs, which primarily provide reactive power support in a grid-following mode, the E-STATCOM extends this functionality by dynamically injecting and absorbing active power through energy storage systems such as supercapacitors (SCs), electrochemical batteries, or flywheels. The letter “E” in E-STATCOM represents “energy”, emphasizing its additional capability to support frequency stability alongside dynamic voltage control. While traditional STATCOMs enhance voltage stability, they do not contribute to inertial response, making them less effective in power systems with high renewable energy penetration. In contrast, an E-STATCOM with grid-forming (GFM) control can emulate the behavior of an SG, providing both inertial response and fast frequency regulation, making it a valuable asset for modern power grids.

Transmission system operators (TSOs) have recognized E-STATCOMs as a promising solution for short-term load fluctuation compensation and inertial response provision. In December 2020, the four German TSOs outlined the need for grid-forming (GFM) E-STATCOM plants [8]. In 2022, a ± 300 Mvar E-STATCOM was commissioned in Germany as a pilot project, delivering inertia support for 10 s. Other initiatives, such as the ViSync project in Spain, aim to integrate SCs and Li-ion batteries into an E-STATCOM for high-voltage transmission grid support in the Lanzarote–Fuerteventura system (Canary Islands, Spain) [9]. Similar developments are underway in Australia [10] and Hawaii, USA [11]. Recently, the E-STATCOM has emerged as a promising alternative solution to compensate for wind power plants [12], offering black-start capability and meeting islanded operation requirements, among other services.

Beyond energy storage integration, the E-STATCOM’s distinguishing feature is its grid-forming (GFM) control strategy. Unlike grid-following converters, which require a stable external voltage reference, GFM converters actively regulate voltage and frequency, making them crucial for weak grids and isolated systems. These devices are designed to maintain a constant or nearly constant internal voltage phasor within sub-transient to transient time frames [13]. An extensive literature on the control of GFM converters is available, with a good summary provided in references [14,15]. According to their principle of operation, GFM converters can be classified into three categories: (a) droop control, (b) synchronous-machine-based control, and (c) other control methodologies such as virtual oscillator-based methods [16], virtual-flux orientation [17,18], reactive power synchronization [19,20], etc.

An essential aspect of GFM operation is current limitation during sudden voltage drops or frequency changes. Several methods have been developed, including virtual impedances [21,22], virtual admittances [23], and nested voltage–current controllers [24]. These techniques are now being widely studied to meet TSO-DSO requirements globally [25–30]. Furthermore, some recent grid codes have started specifying explicit requirements for E-STATCOMs, ensuring their seamless integration into modern power networks [31].

Regarding GFM modeling, either electromagnetic transient (EMT) models or phasor-based RMS models can be used. EMT models provide time-variant instantaneous values of voltage and current, offering high accuracy but at a significant computational cost, making them impractical for large-scale power system simulations. In contrast, RMS models consider only the fundamental frequency components, providing a more computationally efficient alternative while maintaining sufficient accuracy for stability studies. Traditionally, power system analysis neglects transmission line dynamics, relying on algebraic models

that express the relationships between bus voltages and line currents. Reference [32] compares EMT and RMS phasor models for grid-forming converters, analyzing different control strategies and their effects on fast and slow synchronization dynamics. In [33], the suitability of phasor-approximation models for GFM converter simulations is examined, identifying specific conditions for outer-loop control gains that allow TSOs to effectively use RMS models in stability assessments.

The choice of an RMS-based phasorial model over an EMT representation is motivated by the trade-off between computational efficiency and modeling precision. While EMT simulations capture fast-switching transients and harmonic interactions with high accuracy, they require extremely small time steps and high computational resources, making them impractical for large-scale system stability studies. In contrast, RMS models effectively capture the fundamental frequency dynamics of the power system while significantly reducing computational effort. Given that the focus of this study is on evaluating the system-wide stability impact of an E-STATCOM with grid-forming (GFM) control, rather than the detailed switching behavior of power electronic components, the RMS approach is the most suitable choice for this analysis.

This paper proposes a GFM-RMS E-STATCOM model incorporating technical aspects not considered in the consulted bibliography:

- It presents a governor model with an internal power system stabilizer (PSS), as proposed in [17], designed to suppress active power oscillations during setpoint changes or frequency disturbances, particularly when a high inertia constant is used. This model integrates an active current limiter (ACL) to prevent overcurrent excursions resulting from sudden frequency shifts or phase jumps, ensuring the stability of the RMS-based GFM converter. Additionally, an exciter model determines the GFM internal voltage magnitude based on DC-voltage dynamics, incorporating a nonlinear modulation function. Similar to the governor model, a reactive current limiter (RCL) is included for voltage magnitude regulation, enhancing the converter's response under grid disturbances.
- It incorporates a conventional supercapacitor (SC) model, but introduces a novel control strategy that enables DC-voltage variations in response to active power exchanges between the E-STATCOM and the grid, a feature specifically developed in this work. Since RMS-based models focus on fundamental frequency dynamics, most grid-forming models in the literature assume a constant DC bus, typically modeled as an ideal DC source. However, this assumption does not accurately represent real-world conditions, where energy storage systems (ESSs) cannot instantaneously supply power. By dynamically adjusting the DC voltage according to power exchanges, the proposed RMS-based control strategy ensures a more accurate representation of energy storage behavior in E-STATCOM applications, while maintaining computational efficiency for large-scale stability studies.

The main contributions of this work are as follows:

- Development of a complete RMS-based GFM E-STATCOM model specifically designed for power system stability studies. The key innovation of this work lies in the integration of the governor with an internal power system stabilizer (PSS) and an energy storage system (ESS), all within a control architecture tailored for an RMS framework. This model provides a computationally efficient yet accurate representation of grid-forming dynamics, making it well suited for large-scale stability analyses in weak and isolated grids.
- Integration of a nonlinear modulation function in the exciter model, enhancing the representation of real-world modulation techniques used in power electronic converters. While electromagnetic transient (EMT) models explicitly capture high-frequency

switching harmonics, this study employs an RMS phasorial model, focusing on fundamental frequency dynamics relevant to power system stability studies. The inclusion of nonlinear modulation in the RMS framework is justified as it captures the nonlinear relationship between internal voltage magnitude and DC-link voltage, ensuring a realistic dynamic response without requiring high-resolution EMT simulations. Since RMS models inherently do not capture harmonics, any potential distortion effects introduced by modulation do not manifest in this formulation, making harmonic concerns irrelevant in the context of RMS-based stability studies.

- Enhanced damping of power oscillations is achieved through the integration of a power system stabilizer (PSS) within the governor model, even in systems with a high damping coefficient. This approach effectively suppresses oscillatory power modes, a challenge often underestimated in low-inertia grids, thereby improving frequency stability during sudden disturbances. Additionally, an active current limiter (ACL) is incorporated to independently regulate active and reactive current, ensuring robust performance and preventing excessive phase shifts even under large disturbances.
- A comprehensive stability assessment of an isolated power system, using the Fuerteventura–Lanzarote grid (Canary Islands) as a case study. The system's size, topology, and low inertia make it highly susceptible to large frequency deviations and system instabilities under disturbances. A comparative study is conducted, analyzing the responses of an RMS-based GFM E-STATCOM and a SynCon, demonstrating their respective impacts on frequency regulation, voltage support, and fault response.

The paper is organized as follows: Section 2 presents the description of the E-STATCOM system, including the algebraic equations governing the GFM currents and voltage components in different reference frames. Section 3 outlines the governor and exciter models of the GFM converter, incorporating active and reactive current limiters to ensure stable operation. Section 4 develops the dynamic equations of the synchronous condenser (SynCon) using power and energy parameters for a clearer representation of its behavior. Section 5 presents the simulation results, analyzing the impact of the proposed E-STATCOM model on the stability of the Fuerteventura–Lanzarote (Canary Islands, Spain) power system, including comparisons with a SynCon under different operating scenarios. Finally, Section 6 summarizes the conclusions and outlines potential future research directions.

2. RMS-Based GFM E-STATCOM System Description

Figure 1 illustrates an RMS-based GFM E-STATCOM scheme composed of a generic energy storage system (ESS) linked to a grid-forming converter (GFM) on the DC side, which is connected to an AC electric network through an impedance $Z_G e^{j\varphi}$. At the point of common coupling (PCC), the voltage vector is represented by $V_G e^{j\theta}$, where V_G is the voltage magnitude and θ is its angle with respect to an infinite bus $V_\infty \angle 0^\circ$. The current injected into the line is denoted as $I_G e^{j\gamma}$, and the internal voltage of the converter is represented as $E e^{j\delta}$. The ESS shown in Figure 1 is made of a supercapacitor (SC) bank. The voltage and the current on the DC bus are denoted as V_{dc} and I_{dc} .

The control system of the RMS-based GFM E-STATCOM is organized into five blocks, as depicted in Figure 2. The GFM block takes the voltage vector $V_G e^{j\theta}$ at the AC terminals of the converter and the internal voltage vector $E e^{j\delta}$ as inputs. The modulus E is obtained from the exciter model, and the angle δ is from the governor model. By calculating the difference between these voltage vectors and considering the output filter impedance of the GFM converter, the active (I_{aG}) and reactive (I_{rG}) components of the AC current are determined. Additionally, I_{dc} is computed as a function of the AC current components and the amplitude modulation index, m_a , which serves as the control variable for the exciter

model, as explained below. The GFM model also establishes the maximum values for the active current (I_{aG}^{max}) and reactive current (I_{rG}^{max}) based on the operating point of the GFM converter.

E-STATCOM

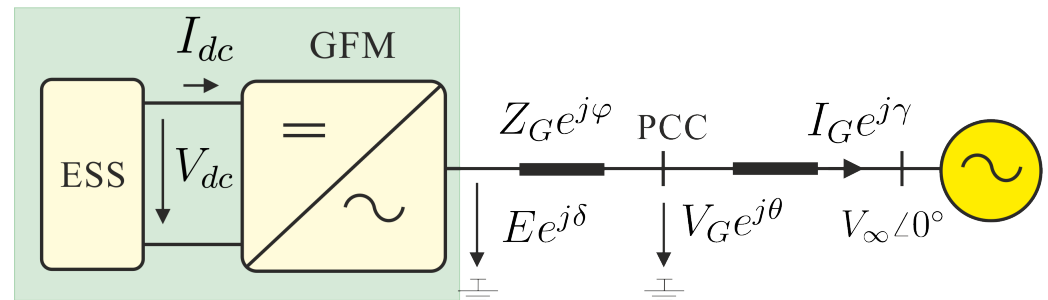


Figure 1. System description of a grid-connected RMS-based GFM E-STATCOM.

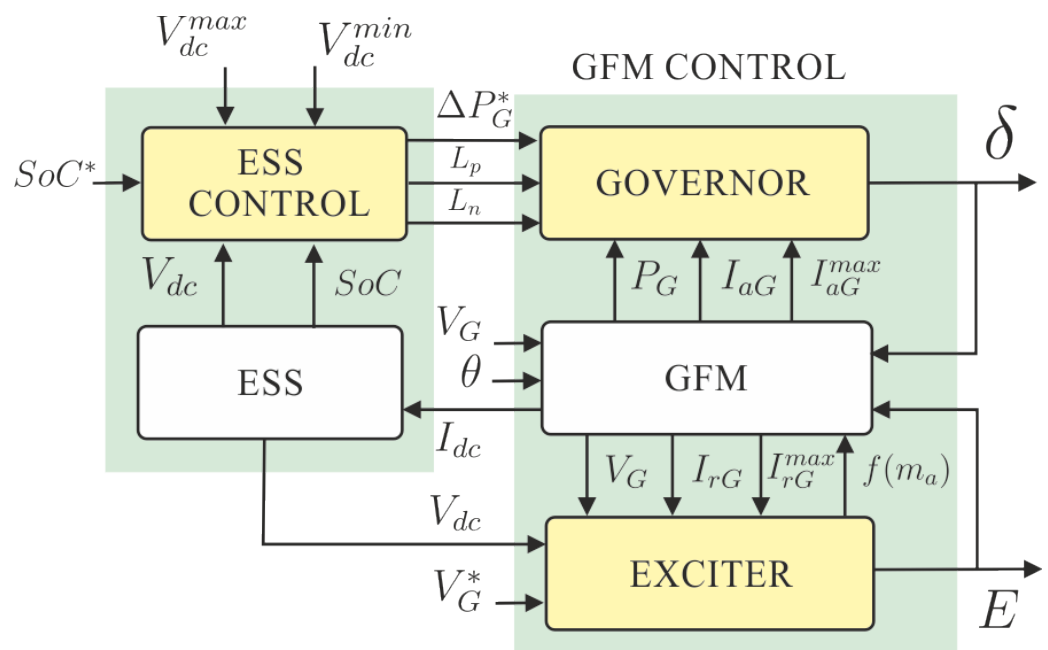


Figure 2. RMS-based GFM E-STATCOM control block diagram.

The governor model computes the angle δ of the internal voltage using the active power reference P_G^* (obtained from the ESS control block) and the calculated active power P_G from the GFM model. This model incorporates an active current limiter (ACL) that limits the angle δ . The ACL's operation mode depends on the active current I_{aG} , its maximum value I_{aG}^{max} , and the limits established by the ESS control block (L_p and L_n). This occurs when the voltage V_{dc} reaches any of its operating limits, V_{dc}^{max} and V_{dc}^{min} . Similarly, the exciter model determines the magnitude E of the internal voltage to regulate V_G at the GFM converter terminals. This magnitude is influenced by the V_{dc} voltage and a nonlinear function of the amplitude modulation index m_a . Similar to the governor model, the exciter model includes a reactive current limiter (RCL) acting on m_a . In this case, the RCL's operation mode depends on the reactive current I_{rG} and its maximum value I_{rG}^{max} .

The ESS model provides the output V_{dc} , calculated as a function of I_{dc} . By integrating I_{dc} , the state of charge (SoC) of the storage system is determined. The ESS control block sets the active power reference P_G^* for the governor based on a specified SoC^* reference. It also establishes limits for the ACL when V_{dc} approaches its constraints.

3. GFM Converter Modeling and Control

The comprehensive modeling and control framework of the GFM converter, encompassing both the governor and exciter models, is elaborated upon in the subsequent sections.

3.1. GFM Model

A vector diagram of the AC voltages and currents of the GFM model is presented in Figure 3a. The DQ axes are considered as the global reference frame rotating at the fundamental frequency ω_0 , with axis D as the angular reference. The internal voltage angle δ defines the q -axis of another reference frame rotating at frequency ω , aligning with the vector $\vec{E} = Ee^{j\delta}$. When the GFM converter is synchronized to the grid, both frequencies, ω_0 and ω , are the same.

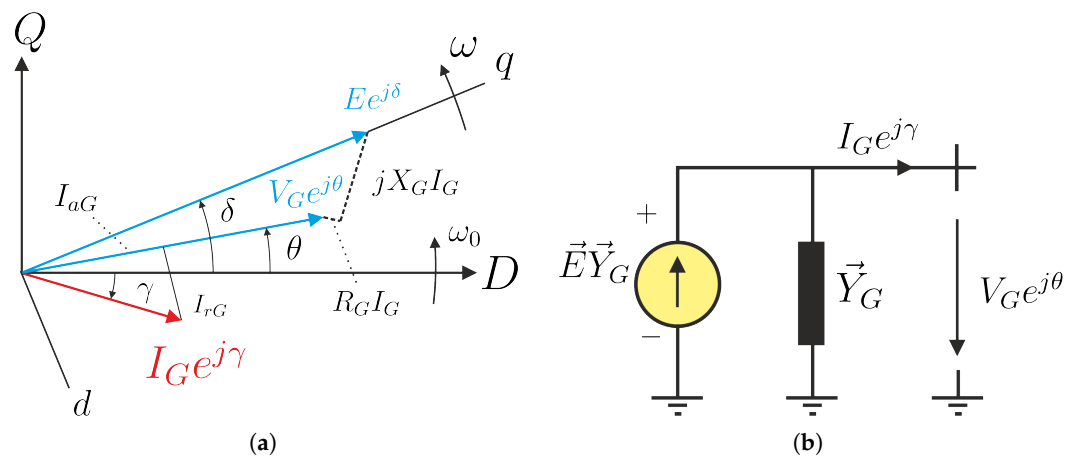


Figure 3. Vector diagram and Norton equivalent of the GFM converter. (a) Vector diagram. (b) Norton equivalent.

The GFM acts as a real voltage source, where its magnitude E and angle δ are variable states of the GFM model. The GFM output filter dynamics is neglected to ensure compatibility with the transmission network so that it is represented as a complex impedance $Z_G e^{j\varphi} = R_G + jX_G$, the inverse of which is the admittance $\vec{Y} = G_G + jB_G$. The current vector injected into the grid is determined through the Norton equivalent, as depicted in Figure 3b and expressed in Equation (1),

$$I_{DG} + jI_{QG} = I_G e^{j\gamma} = (G_G + jB_G)(Ee^{j\delta} - V_G e^{j\theta}) \quad (1)$$

and separated into real and imaginary parts in Equations (2) and (3):

$$I_{DG} = G_G(E \cos \delta - V_G \cos \theta) - B_G(E \sin \delta - V_G \sin \theta) \quad (2)$$

$$I_{QG} = G_G(E \sin \delta - V_G \sin \theta) + B_G(E \cos \delta - V_G \cos \theta) \quad (3)$$

When the current vector $I_{DG} + jI_{QG}$ is expressed in a reference frame aligned to the voltage $V_G e^{j\theta}$, the active and reactive components are obtained as follows,

$$I_{aG} + jI_{rG} = (I_{DG} + jI_{QG})e^{-j\theta} = (G_G + jB_G)(Ee^{j(\delta-\theta)} - V_G) \quad (4)$$

and this is separated into real and imaginary parts in Equations (5) and (6):

$$I_{aG} = G_G(E \cos (\delta - \theta) - V_G) - B_GE \sin (\delta - \theta) \quad (5)$$

$$I_{rG} = G_GE \sin (\delta - \theta) + B_G(E \cos (\delta - \theta) - V_G) \quad (6)$$

These two components, I_{aG} and I_{rG} , are derived separately to ensure that each can be limited independently, thereby ensuring that the current does not surpass the maximum limit set for the GFM. Active and reactive power are then directly calculated in per unit (p.u.) using Equations (5) and (6), as per Equation (7),

$$P_G + jQ_G = V_G(I_{aG} - jI_{rG}) \quad (7)$$

so that, active power P_G is proportional to the active current I_{aG} and the voltage magnitude V_G . Similarly, the reactive power Q_G is given by the product of V_G and the reactive current I_{rG} , but with a minus sign. When the complex impedance $Z_G e^{j\varphi}$ has only an imaginary part, $G_G = 0$ and $B_G = -1/X_G$; substituting these values of the complex admittance into Equations (5) and (6) yields the well-known expressions for active and reactive power, with all variables in p.u. [34], as given by Equations (8) and (9).

$$P_G = \frac{V_G}{X_G} E \sin(\delta - \theta) \quad (8)$$

$$Q_G = \frac{V_G}{X_G} (E \cos(\delta - \theta) - V_G) \quad (9)$$

As shown in Figure 2, the GFM model provides the active and reactive current limits to the control blocks. In general, for a GFM operating as a static synchronous compensator, these limits are variable to optimize its performance. For instance, when there is no frequency change in the grid, the GFM does not exchange active power, and its active current limit I_{aG}^{max} could be zero. All the current capacity of the converter I_G^{max} is used to exchange reactive power for voltage control, so that $I_{rG}^{max} = I_G^{max}$. On the contrary, when no reactive power is required, all the current capacity could also be active, and $I_{aG}^{max} = I_G^{max}$. However, it is interesting to allocate part of the current for voltage control since the frequency response can last several seconds or even minutes, and therefore I_{rG}^{max} should be nonzero. Thus, the maximum reactive current I_{rG}^{max} can be calculated considering the total maximum current I_G^{max} and the actual active current I_{aG} , as given in Equation (10).

$$I_{rG}^{max} = \sqrt{(I_G^{max})^2 - (I_{aG})^2} \quad (10)$$

In this scenario, I_{aG} is constrained by its maximum value I_{aG}^{max} , which must be less than I_G^{max} .

3.2. Exciter Model

The exciter model calculates the internal voltage magnitude E based on the voltage reference V_G^* and its actual value V_G . Within this block, a controller is employed, and its output is the amplitude modulation index, denoted as m_a . The index m_a is defined as the relationship between the internal peak voltage per phase of the GFM, $\hat{V}_{ac}$, and half of the DC voltage, $V_{DC}/2$ [35]. If, as usual, the voltage bases for AC and DC are $\hat{V}_{ac,b}$ and $V_{DC,b} = 2\hat{V}_{ac,b}$, respectively, the index m_a can be expressed as given by Equation (11):

$$m_a = \frac{\hat{V}_{ac}}{V_{DC}/2} = \frac{E}{V_{dc}} \quad (11)$$

where V_{DC} represents the DC voltage in volts, and V_{dc} is its value in p.u. Likewise, E is the internal voltage of the GFM in p.u.

Figure 4 illustrates the nonlinear relationship between $\frac{E}{V_{dc}}$ and m_a , depicting three distinct zones: (a) linear modulation when $m_a < 1$; (b) overmodulation, where the voltage magnitude E does not increase proportionally with $m_a \cdot V_{dc}$; and (c) square-wave operation,

characterized by $\frac{E}{V_{dc}} = \frac{4}{\pi}$, meaning the internal maximum voltage magnitude is $1.27 \cdot V_{dc}$ in p.u. As per [35], square-wave operation is attained when $m_a > 2.16$ for a frequency modulation index m_f of 20.

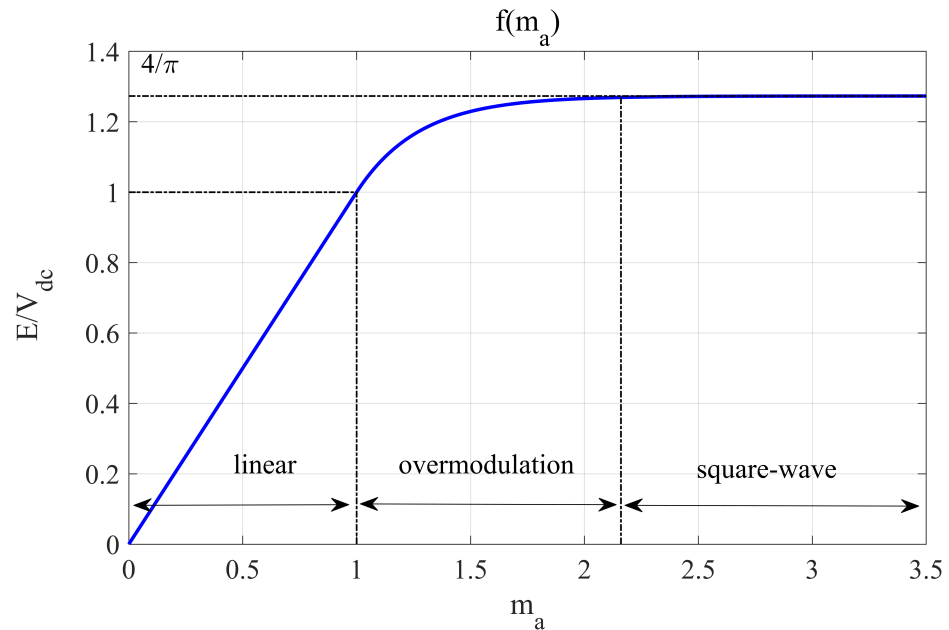


Figure 4. Nonlinear relationship between $\frac{E}{V_{dc}}$ and m_a .

The nonlinear modulation function, $f(m_a) = \frac{E}{V_{dc}}$, is represented by Equation (12):

$$\begin{aligned} f(m_a) &= m_a \quad 0 < m_a < 1 \\ f(m_a) &= 1 + A \left(1 - e^{-\frac{(m_a-1)}{A}} \right) \quad m_a \geq 1 \end{aligned} \quad (12)$$

where A is a constant equal to $\left(\frac{4}{\pi}\right) - 1$.

One concern with nonlinear modulation is the potential for current harmonic distortion, which grid codes typically restrict for transmission-connected devices [31]. However, in the context of an RMS model, harmonics are not explicitly represented, as the model only captures fundamental frequency dynamics. The function $f(m_a)$ used in this work reflects the nonlinear relationship between the internal voltage magnitude and the DC-link voltage without introducing harmonic content in the RMS formulation. Since the RMS approach inherently filters out switching harmonics, concerns about harmonic distortion in the overmodulation region are not relevant within this modeling framework.

According to the previous section, the GFM model comprises a set of nonlinear expressions for the vector $I_G e^{j\gamma}$, where its DQ current components are indicated in Equations (2) and (3), and its active and reactive components are outlined in Equations (5) and (6). Additionally, as mentioned earlier, expressions for active and reactive power, crucial for GFM control, are presented in Equation (7). In Figure 2, it is evident that I_{dc} is also an output of the GFM model, calculated by performing an active power balance (neglecting converter losses) on the AC and DC terminals of the GFM. The AC active power expression, using dq components aligned to the internal voltage vector $E e^{j\delta}$, is given by $E I_q$, so that

$$V_{dc} I_{dc} = E I_q \quad (13)$$

dividing both terms by V_{dc} and using the modulation function of Equation (12), the definition I_{dc} is given by Equation (14),

$$I_{dc} = f(m_a)I_q \quad (14)$$

where I_q is calculated by rotating the current vector $I_{DG} + jI_{QG}$ to the dq reference frame, as is given by Equation (15),

$$I_d + jI_q = (I_{DG} + jI_{QG})e^{-j(\delta - \frac{\pi}{2})} \quad (15)$$

so, by separating into real and imaginary parts and substituting I_q into Equation (14), I_{dc} can be expressed as Equation (16):

$$I_{dc} = f(m_a)(I_{DG} \cos \delta + I_{QG} \sin \delta) \quad (16)$$

Similarly, the internal voltage magnitude E is determined according to Equation (12) as

$$E = f(m_a)V_{dc} \quad (17)$$

Figure 5 depicts the excitation model of the GFM. This model aims to determine the internal voltage magnitude E of the GFM, with its reference E^* calculated according to Equation (17) as the product of V_{dc} and the modulation function $f(m_a)$. To account for converter dynamics, a first-order function relating E and E^* has been employed. The modulation index, m_a , is derived from a voltage droop regulator, as is shown in Equation (18).

$$m_a = m_{a0} - n_q(V_G - V_G^*) + \Delta m_a \quad (18)$$

The first term, m_{a0} , represents a constant value. The second is the droop term, $-n_q\Delta V_G$, which has a negative sign and is proportional to the voltage increment $\Delta V_G = V_G - V_G^*$. The control signal Δm_a modifies the modulation index if the reactive current I_{rG} exceeds any of its limits; otherwise, its value remains zero. The formulation of Equation (18) is based on previously established methodologies for voltage control in grid-forming (GFM) converters. Specifically, similar expressions have been proposed in the literature to regulate the internal voltage magnitude based on system conditions and modulation strategies, as discussed in [15]. This approach is fundamental in ensuring stable voltage reference generation in weak and isolated grids.

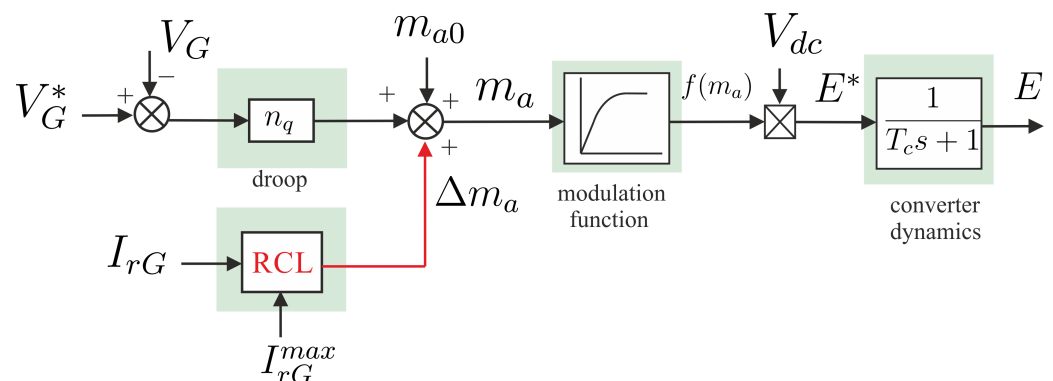


Figure 5. GFM exciter model.

Equation (18) can be interpreted as follows: when the voltage V_G exceeds its reference value V_G^* (i.e., $V_G > V_G^*$), E should be decreased to reduce the generated reactive power and consequently lower the voltage V_G . Conversely, when V_G is lower than its reference ($V_G < V_G^*$), E should be increased. The reactive current limiter (RCL) block, illustrated in Figure 5, computes Δm_a based on I_{rG} and I_{rG}^{max} . Figure 6 shows the RCL logic: when

I_{rG} exceeds I_{rG}^{max} , a proportional regulator, K_q , generates a negative Δm_a to reduce the internal voltage magnitude E , thereby limiting I_{rG} . If the reactive current remains within limits, such as when $I_{rG} < I_{rG}^{max}$, the regulator output saturates at $\Delta m_a = 0$. A symmetrical regulator is employed for negative values of I_{rG} .

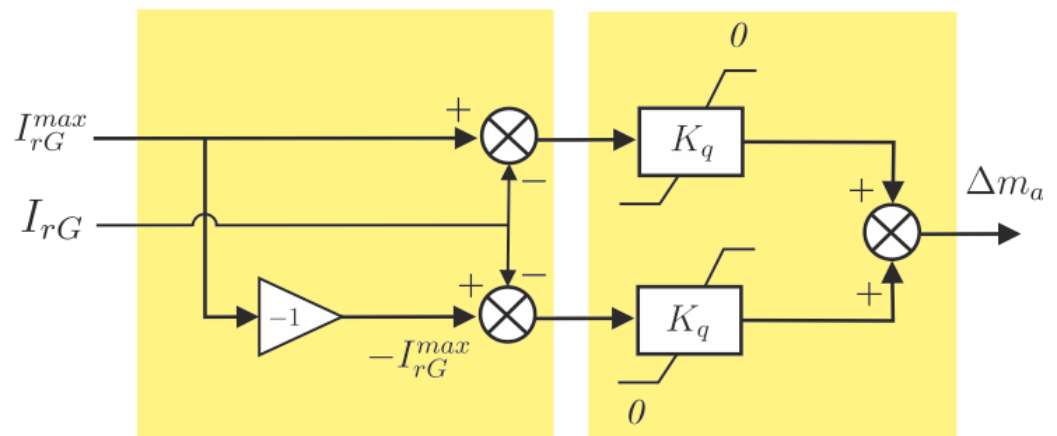


Figure 6. Reactive current limiter (RCL) block.

3.3. Governor Model

The governor model refers to the active power synchronization loop used in GFM converters to emulate the swing equation of an SG. As shown in Figure 2, this block takes the actual power P_G and its reference P_G^* as inputs, along with the active current I_{aG} and its maximum limit I_{aG}^{max} . The output of this block is the power angle δ , representing the angular position of the internal voltage vector $Ee^{j\delta}$ with respect to the DQ stationary reference frame (see Figure 3a). The dynamic equation for δ is obtained by integrating the rotational speed ω with respect to the synchronously rotating reference ω_0 in rad/s [34], as given by Equation (19):

$$\frac{1}{\omega_0} \frac{d\delta}{dt} = \Delta\omega \quad (19)$$

where $\Delta\omega$ is the difference between ω and ω_0 in p.u.

As can be seen in Figure 7, the frequency increment $\Delta\omega$ is composed of the sum of three components. The first component, ω_1 , is the output of the first-order block, representing the swing equation, defined by the inertia constant H and damping factor D . The dynamic of ω_1 is given by Equation (20), which is derived from the classical power swing equation used to model frequency behavior in electrical power systems. This formulation has been extensively studied in the context of synchronous generators and grid-forming converters, particularly in low-inertia systems, where its influence on frequency stability is crucial [2,34]. The equation captures the interplay between system inertia, damping, and power imbalances, making it fundamental for maintaining stable frequency control.

$$2H \frac{d\omega_1}{dt} + D\omega_1 = P_G^* - P_G \quad (20)$$

Both parameters, H and D , indicate the active power variation of the GFM during a frequency event to ensure its synchronization. In the case of the inertia constant (H), the GFM converter exchanges extra active power only when the frequency is changing, while the damping factor (D) comes into play when a frequency deviation occurs. One of the key advantages of GFM over real SGs is that H and D can be readily changed during operation, with the only restriction being having sufficient energy on the DC bus to provide the required inertia and frequency response. However, it is not always possible to choose any set of H and D parameters since, in some cases, the active power oscillation can

become unacceptable [17]. For example, when designing a GFM with only inertial response ($D = 0$), the transfer function of the active power under a frequency change presents two complex conjugate poles on the imaginary axis, representing a critically stable system with an undamped oscillatory mode.

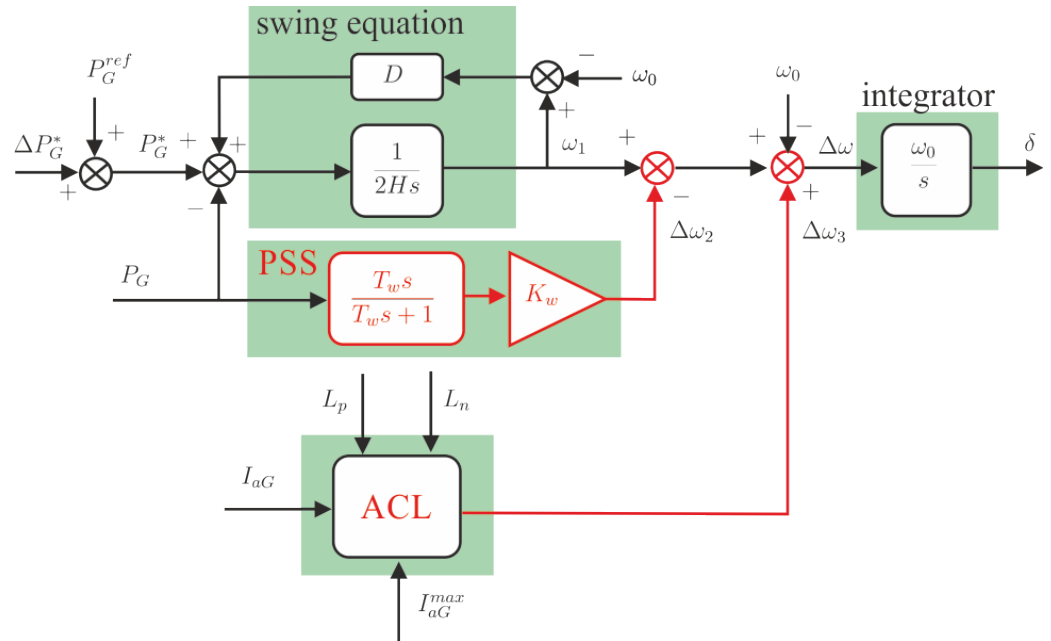


Figure 7. GFM governor model.

As depicted in Figure 7, a new component, $\Delta\omega_2$, is added to the frequency ω_1 . This signal is derived from the PSS block, where the actual active power P_G is filtered through a washout filter with a time constant T_w and a gain constant K_w . Consequently, the output signal $\Delta\omega_2$ is only non-null during changes in P_G and is zero in steady state. This signal has a negative sign, indicating that the PSS counteracts active power changes. The dynamic equations of the PSS block are given by Equations (21) and (22).

$$T_w \frac{d\Delta z}{dt} = P_G - \Delta z \quad (21)$$

$$\Delta\omega_2 = K_w(P_G - \Delta z) \quad (22)$$

where Δz is an auxiliary variable.

While conventional grid-forming control strategies already incorporate a high damping coefficient (D), the addition of a power system stabilizer (PSS) is justified in cases where synthetic inertia effects introduce low-frequency oscillation modes [2]. These oscillations are particularly relevant in weak grids, where sudden disturbances can lead to prolonged transient instabilities. The PSS proposed in this work enhances power oscillation damping, ensuring faster frequency stabilization after large power imbalances.

Similarly, as indicated in Figure 7, another component in the frequency increment, $\Delta\omega_3$, is introduced in the active power synchronization loop to limit the active current. The control logic for this active current limiter (ACL) block is shown in Figure 8. The maximum current limit, I_{aG}^{max} , is multiplied by L_p and L_n , two inputs derived from the ESS control block. L_p is set to 1 when the DC voltage, V_{dc} , is within limits, and 0 if $V_{dc} \leq V_{dc,min}$, preventing further discharge of the ESS to avoid a reduction in V_{dc} . Conversely, L_n is set to -1 when V_{dc} is within limits, and 0 if $V_{dc} \geq V_{dc,max}$. When this condition is met, the ESS charging stops, and V_{dc} is limited to its maximum value, $V_{dc,max}$.

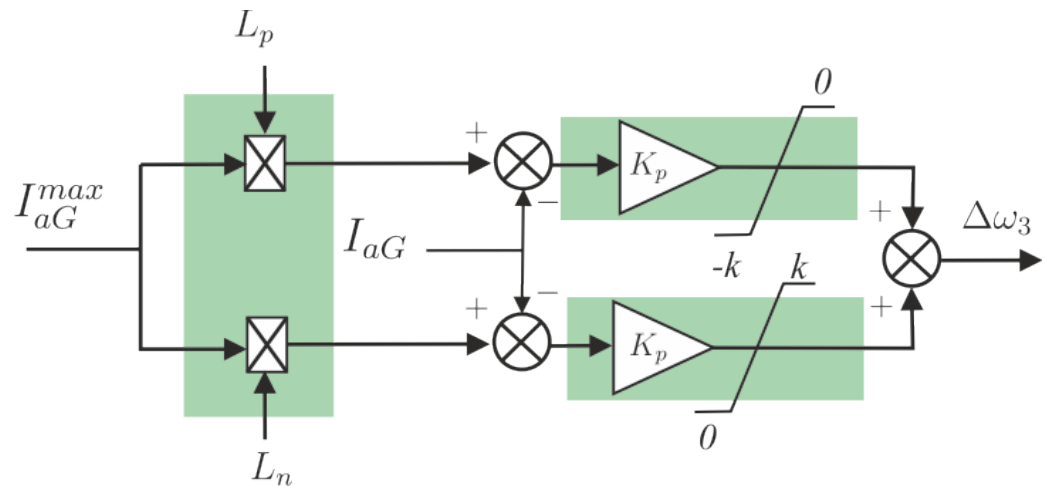


Figure 8. Active current limiter (ACL) block.

As shown in Figure 8, if $-I_{aG}^{max} < I_{aG} < I_{aG}^{max}$, both upper and lower limiters saturate to zero, and therefore $\Delta\omega_3 = 0$. However, when $I_{aG} > I_{aG}^{max}$, the upper limiter saturates to $-k$, and the lower limiter to 0, so that $\Delta\omega_3 = -k$, i.e., the power angle δ is reduced, and therefore the active current is limited. On the contrary, when $I_{aG} < -I_{aG}^{max}$, the upper limiter saturates to 0, and the lower limiter to k , so that $\Delta\omega_3 = k$. In this case, the power angle is negative, and its absolute value is reduced. Finally, K_p is a gain that determines the time response to reach the saturation limits when I_{aG} exceeds I_{aG}^{max} or $-I_{aG}^{max}$. The higher K_p is, the shorter the time response.

One of the challenges of separate active and reactive current limitations is ensuring robustness under large phase shifts, where a simple proportional control K_p may not be sufficient. To address this issue, the active current limiter (ACL) in this work dynamically adjusts $\Delta\omega_3$ based on the ESS state and the active current limits I_{aG}^{max} . This prevents excessive phase shifts that could destabilize the synchronization process. The simulation results in Section 5 confirm that the proposed ACL effectively limits current overshoots while maintaining stable power exchange, even under significant disturbances.

The ACL/RCL control logic operates by sequentially verifying the active and reactive current limits to adjust the power angle (δ) and modulation index (m_a). The process begins by checking the active current (I_{aG}). If it exceeds the predefined limit (I_{aG}^{max}), the power angle δ is adjusted accordingly to regulate the current. Otherwise, δ remains unchanged. Next, the reactive current (I_{rG}) is evaluated. If it surpasses its threshold (I_{rG}^{max}), the modulation index (m_a) is modified to bring the current within acceptable levels. If the reactive current is within limits, m_a remains unchanged. Finally, the updated values of δ and m_a are applied to the converter control to ensure proper operation.

4. Energy Storage System (ESS) Modeling and Control

Usually in grid studies, when considering frequency and inertia response through IBRs, it is a common practice to assume an ideal DC-voltage source connected to the converter DC bus, implying that the converter and the DC-voltage source can supply all the active power required in the event of a change in the grid frequency, with the only limitation being to not exceed the maximum current of the converter. However, this assumption is not realistic because the ESS capacity is always limited, and its behavior affects the DC-voltage level, consequently influencing the IBRs' dynamics. In this work, an ESS connected at the DC terminals based on an SC has been considered.

4.1. Supercapacitor Model

Supercapacitors (SCs) are energy storage devices that offer high power density but lower energy density compared to batteries. Due to their ability to deliver stored energy rapidly and withstand tens of thousands of charge/discharge cycles, they are well suited for power applications in energy storage systems (ESSs). Figure 9 shows the conventional equivalent circuit model of an SC, which consists of an equivalent resistor R_s connected in series with a capacitor C_{sc} . Additionally, a parallel resistor R_p is included at the capacitor terminals to represent the SC's self-discharge.

This work introduces a novel control strategy that dynamically adjusts the DC voltage in response to active power exchanges between the E-STATCOM and the grid. While most RMS-based grid-forming models assume a constant DC bus, typically modeled as an ideal DC source, this assumption does not accurately reflect real-world conditions, where energy storage systems (ESSs) exhibit dynamic behavior. By integrating a control mechanism that allows the DC voltage to vary according to power exchanges, this approach enhances the accuracy of the SC representation while maintaining computational efficiency for large-scale stability studies.

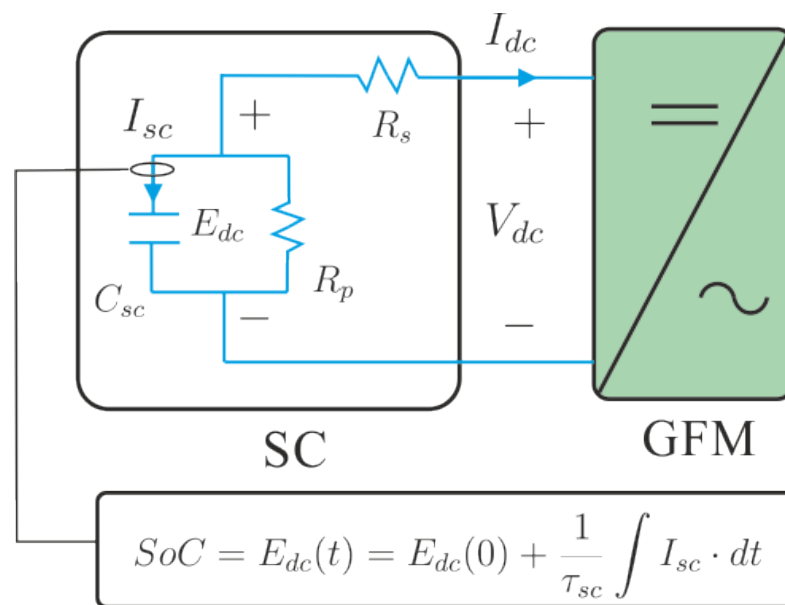


Figure 9. SC equivalent circuit.

The dynamic equation for the SC describes the relationship between the voltage e_{dc} and the current i_{sc} in the capacitor C_{sc} , as given by Equation (23):

$$C_{sc} \frac{de_{dc}}{dt} = i_{sc} \quad (23)$$

Expressing Equation (23) in p.u. and calculating E_{dc} as a function of I_{sc} yields Equation (24),

$$E_{dc}(t) = E_{dc}(0) + \frac{1}{\tau_{sc}} \int I_{sc} dt \quad (24)$$

so that E_{dc} acts as the state of charge (SoC) of the SC, with the SC time constant in Equation (24), τ_{sc} , defined as the product of C_{sc} and the DC impedance base $Z_{b,dc}$.

$$\tau_{sc} = C_{sc} Z_{b,dc} \quad (25)$$

The impedance base $Z_{b,dc}$ is calculated as $V_{DC,b}^2/P_{DC,b}$. It is worth noting τ_{sc} can also be defined as the ratio between the energy (in MJ) and the nominal power (in MW) of the SC.

In Equation (24), I_{sc} is obtained as the sum, with a negative sign, of the current I_{dc} and the self-discharge current E_{sc}/R_p , with R_p in p.u., as given by Equation (26):

$$I_{sc} = -\left(I_{dc} + \frac{E_{sc}}{R_p}\right) \quad (26)$$

Likewise, from I_{dc} and R_s (p.u.), the DC voltage V_{dc} is equal to

$$V_{dc} = E_{dc} - R_s I_{dc} \quad (27)$$

Figure 10 shows the SC block diagram in accordance with the previous equations, where I_{dc} is the input and V_{dc} and SoC are the outputs, all expressed in p.u.

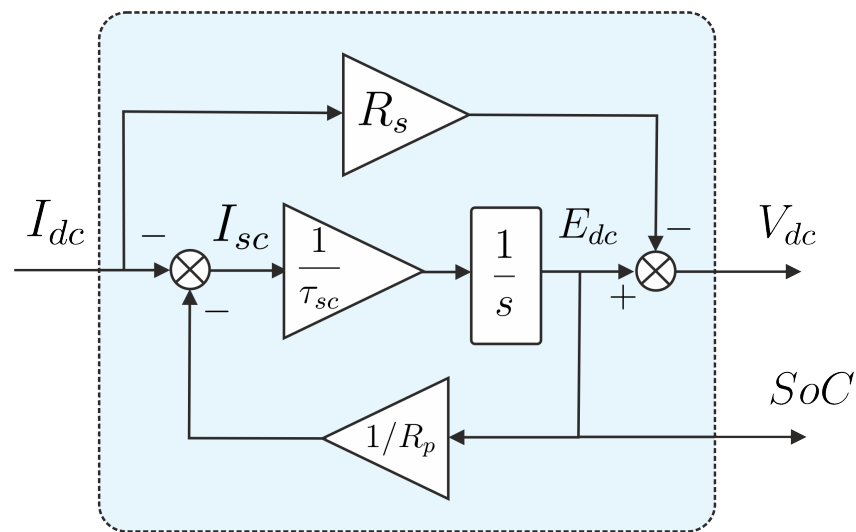


Figure 10. SC block diagram.

4.2. ESS Control

Figure 11 shows the ESS control scheme, whether it involves an SC or a battery. The controlled variable is the state of charge (SoC), which influences the active power reference P_G^* through a proportional–integral (PI) regulator with maximum (L_p) and minimum (L_n) saturation limits. When $V_{dc} < V_{dc,min}$, the maximum saturation limit L_p is set to zero, allowing P_G and I_{aG} to only assume negative values to charge the SC or the battery. Similarly, when $V_{dc} > V_{dc,max}$, indicating a fully charged ESS, the minimum saturation limit L_n is set to zero, preventing charging the battery or SC.

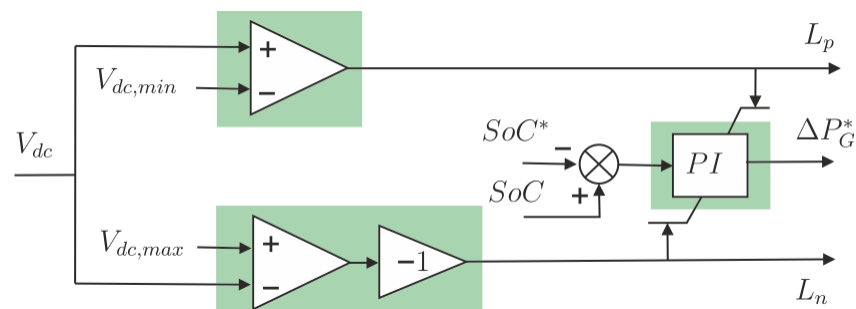


Figure 11. ESS control scheme.

5. Results and Discussion

Simulations are conducted on the power system of the Fuerteventura and Lanzarote islands in the Canary Archipelago, Spain, as shown in Figure 12a. The transmission grid operates at two voltage levels, 132 kV and 66 kV, interconnected by submarine cables linking both islands (shown by dashed lines in Figure 12b). It comprises 17 nodes: 6 at 132 kV, 6 at 66 kV, and 5 generation nodes at 20 kV. The conventional generation includes two synchronous power plants: 265 MVA at Punta Grande on Lanzarote Island, and 187 MVA at Las Salinas on Fuerteventura Island (see Figure 12b). The IBR is aggregated at three transmission nodes, with rated powers of 52 MW (wind and PV at Lanzarote Island), 60 MW (wind generation at Fuerteventura Island), and 44 MW (PV at Lanzarote Island).

Table 1 presents the technical parameters of the Lanzarote–Fuerteventura power grid, including bus connections, the transmission line, and transformer parameters. All parameters are expressed in per unit (p.u.) on a 100 MVA system base. The power grid model is based on the 2026 planning horizon, incorporating two submarine interconnections (66 kV and 132 kV cables) and a 132 kV double-circuit corridor [36].

The dynamic simulations were performed using MATLAB/Simulink to analyze the transient response of the Lanzarote–Fuerteventura power system with the proposed E-STATCOM model. Both the Lanzarote–Fuerteventura power system and the E-STATCOM model were developed within MATLAB/Simulink, ensuring a precise and comprehensive representation of grid dynamics, including inverter-based resources, synchronous condensers, and load interactions. The solver used was ode4 (Runge–Kutta), a solver suitable for power system simulations, with a fixed time step of 100 μ s to accurately capture system dynamics while ensuring numerical stability. The total simulation times were set to 20 and 10 s to cover both the transient and steady-state responses of the system. The model parameters were initialized according to the grid conditions described in Table 1.

Table 1. Transmission line and transformer parameters.

From Bus	To Bus	r	x	b
1	2	0.008744408	0.036143044	0.03491
2	3	0.011673705	0.048250637	0.046604598
3	4	0.00763225	0.031546189	0.030470011
4	5	0.001809211	0.06875	0
5	6	0.006958219	0.022677916	0.022642706
5	8	0.012074674	0.071845088	0.094760385
4	7	0.007593352	0.01328253	0.139133459
7	8	0.001809211	0.06875	0
7	9	0.004480257	0.018518136	0.017886402
9	10	0.001809211	0.06875	0
10	11	0.01929146	0.079820937	0.00481318
10	12	0.031180533	0.10815978	0.001212937
11	12	0.026177686	0.090805785	0.001018324
11	13	0.001413308	0.053705693	0
3	14	0.001817503	0.06906511	0
12	15	0.005553985	0.21105142	0
3	16	0.006620324	0.251572327	0
1	17	0.00485409	0.184455437	0

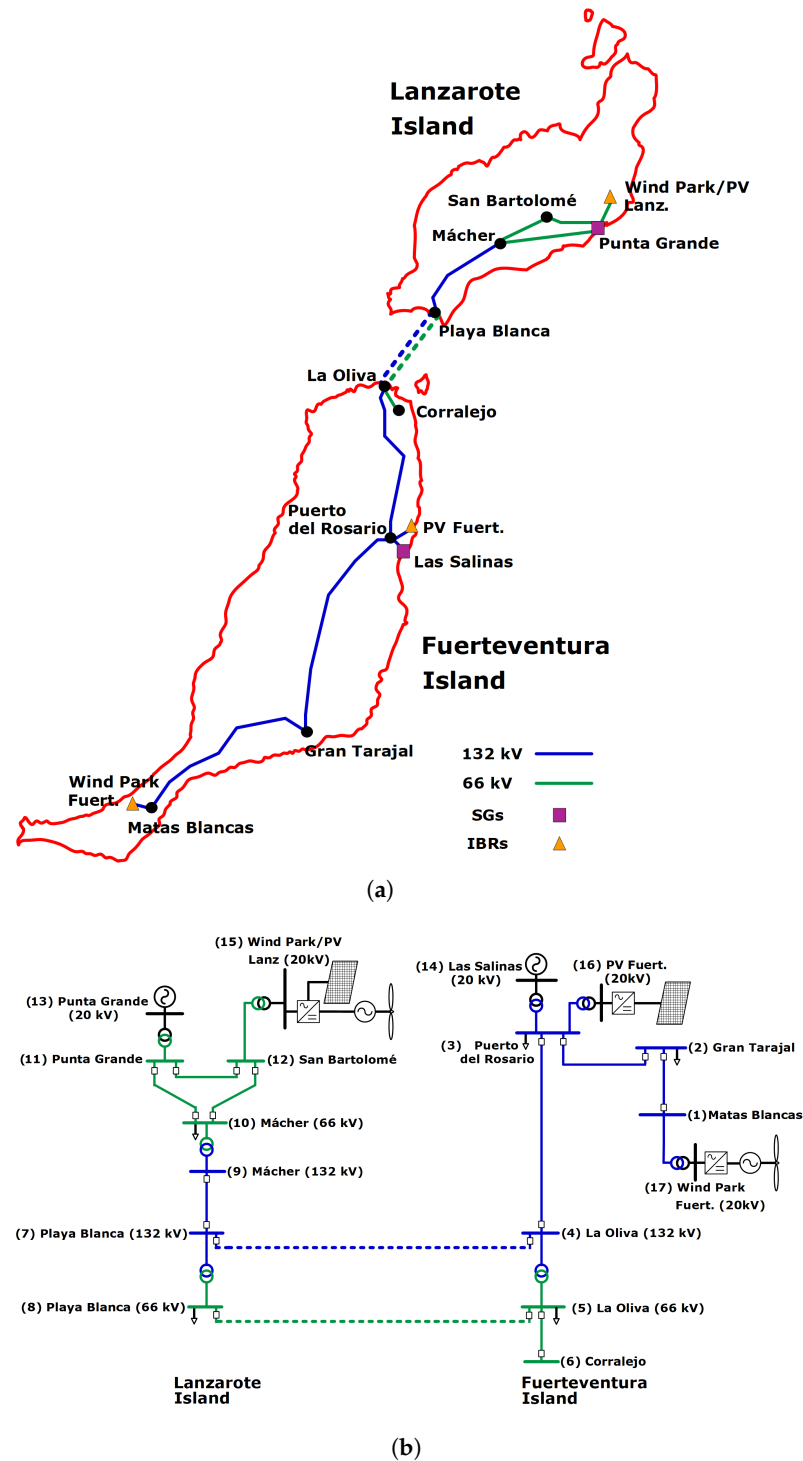


Figure 12. Power grid of Lanzarote and Fuerteventura Islands. (a) Geographical representation of the Lanzarote–Fuerteventura power grid. (b) Single-line diagram of the Lanzarote–Fuerteventura power system.

Two different reference contingencies in the system are simulated to evaluate the response of the E-STATCOM: a synchronous power plant outage and a direct three-phase short circuit. These simulations are conducted across the following three distinct scenarios:

- Case study 1: This scenario represents the base case, the system operates without any compensation device.

- Case study 2: The second scenario involves the connection of a 25 MVA synchronous condenser (SynCon) to the 66 kV Mácher bus (see Figure 12). The SynCon is modeled using the standard “genrou” model [37] with an inertia constant $H = 8$ s.
- Case study 3: The third scenario involves connecting a 25 MVA E-STATCOM with a supercapacitor of 300 MWs to the 66 kV Mácher bus. The E-STATCOM is represented using the proposed model in this work. The parameters of the model are $H = 8$ s, $D = 50$ p.u. (2% droop), $T_w = 0.5$ s, and $K_w = 0.05$ p.u.

The selection of the parameters $H = 8$ s, $D = 50$ p.u., $T_w = 0.5$ s, and $K_w = 0.05$ p.u. was based on established practices for power system stability enhancement. The inertia constant $H = 8$ s was chosen to emulate the response of a synchronous generator, ensuring appropriate frequency stability in the absence of traditional inertia sources [34]. The damping coefficient $D = 50$ p.u. is within the recommended range for isolated grids with high renewable penetration, helping suppress frequency oscillations [2,15]. The washout filter time constant $T_w = 0.5$ s allows effective transient control while preventing steady-state drift [38]. Finally, the power system stabilizer gain $K_w = 0.05$ p.u. ensures sufficient damping without introducing excessive compensation [39].

These parameter values have been validated in previous studies and adapted to the characteristics of the Fuerteventura–Lanzarote grid to achieve an optimal trade-off between stability and responsiveness.

5.1. Synchronous Power Plant Outage

This subsection analyzes the outage of the synchronous power plant at Las Salinas on Fuerteventura Island, Figure 12. At the time of the incident, it was generating 81 MW. Figure 13 shows and compares the evolution of various variables in the system across the three case studies.

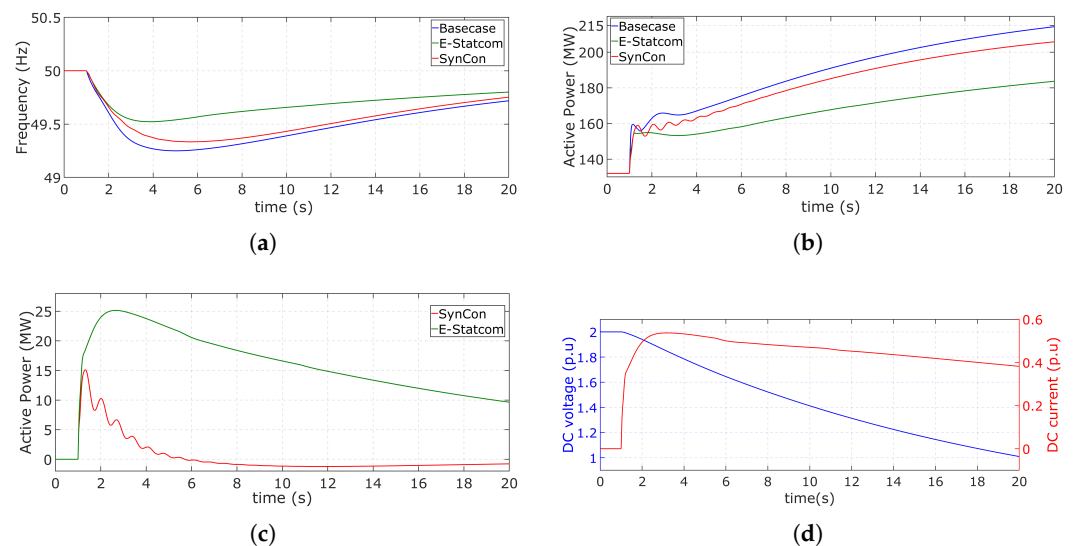


Figure 13. Response to a synchronous power plant outage. (a) Frequency measured at Punta Grande bus. (b) Active power delivered by the Punta Grande power plant. (c) Active power delivered by the synchronous condenser and the E-STATCOM. (d) DC voltage and current at the E-STATCOM DC link.

Figure 13a depicts the system frequency measured at the Punta Grande bus, while Figure 13b shows the active power delivered by the synchronous power plant connected to this bus for the three case studies. In the initial moments after the outage, an inertial response is observed. As shown in Figure 13a, the response during the first second is similar across the three case studies, although slightly better when compensation devices are connected. After the first second, the compensation devices improve the system response,

and help dampen the frequency nadir to higher values. The inertial response is crucial as it temporarily stabilizes the system while the primary frequency regulation takes effect. In all cases, the Punta Grande synchronous power plant responds by supplying power to the grid in order to compensate the imbalance caused by the incident, as illustrated in Figure 13b.

When the SynCon is connected (case study 2), the inertial response improves, and the system stabilizes after reaching a frequency nadir of approximately 49.35 Hz, slightly higher than the base case (49.25 Hz), as shown in Figure 13a. In Figure 13b, it is evident that the initial active power delivered by the Punta Grande SG is lower compared to the base case.

To further quantify the system response, key stability metrics were analyzed. The rate of change of frequency (RoCoF) was computed as the initial frequency slope following the outage. The base case exhibited an RoCoF of 0.26 Hz/s, while the inclusion of a SynCon reduced this to 0.20 Hz/s. The E-STATCOM with GFM demonstrated the best performance, limiting the RoCoF to 0.15 Hz/s, thereby improving the transient frequency stability. The frequency recovery time, defined as the time taken for the frequency to return to within ± 0.1 Hz of nominal, was 10 s for the base case, 8 s for the SynCon, and 6 s for the E-STATCOM, highlighting the faster dynamic response enabled by synthetic inertia and active power injection.

Figure 13c shows the active power delivered by the SynCon, which initially peaks at 15 MW. This power response is dictated by the swing equation of a synchronous machine without a coupled turbine governor, meaning that its power injection is inherently tied to the rate of change of frequency at its connection bus. The fundamental difference in dynamic response between the E-STATCOM with GFM and the SynCon stems from their distinct control mechanisms. The SynCon, governed by its physical inertia, follows the natural swing equation, providing frequency support that is inherently constrained by its mechanical properties. In contrast, the E-STATCOM with GFM actively regulates frequency through synthetic inertia control, enabling a more rapid and sustained injection of active power in response to grid disturbances. This contrast is clearly observed in Figure 13c, where the E-STATCOM maintains active power injection well beyond the initial transient, whereas the SynCon's contribution gradually declines as its inertial effect dissipates.

Upon connecting the E-STATCOM (case study 3), the inertial response improves, and the system stabilizes after it reaches a frequency nadir of 49.48 Hz, which is significantly better than in the other case studies. Another advantage of the E-STATCOM over the SynCon solution is its capacity to maintain primary regulation, as depicted in Figure 13c, as long as the supercapacitor retains energy. Figure 13d shows the DC voltage and current at the supercapacitor, showing how the voltage decreases as the element discharges to stabilize the system. The E-STATCOM has a dual converter system with a total rated power of 2×12.5 MVA and a DC-voltage base of 2200 V for each link.

The energy exchange analysis for grid stabilization highlights a significant difference between the E-STATCOM and the synchronous condenser (SynCon). Over a 20 s period (see Figure 13c), the E-STATCOM delivers approximately 0.060 MWh to the grid, whereas the SynCon supplies only 0.0083 MWh. This energy transfer occurs both during the transient and steady-state responses. A key distinction is that the E-STATCOM sustains its energy delivery over a longer period, as it keeps the DC-link voltage of the VSC relatively constant for an extended duration. While the E-STATCOM provides a rapid dynamic response, its prolonged energy injection could be a crucial factor when assessing long-term operational efficiency.

5.2. Short Circuit in the Grid

This subsection analyses the system response when connecting either a SynCon (case study 2) or an E-STATCOM (case study 3) during a direct three-phase short circuit at La Oliva bus at 66 kV (see Figure 12). The results are presented in Figure 14.

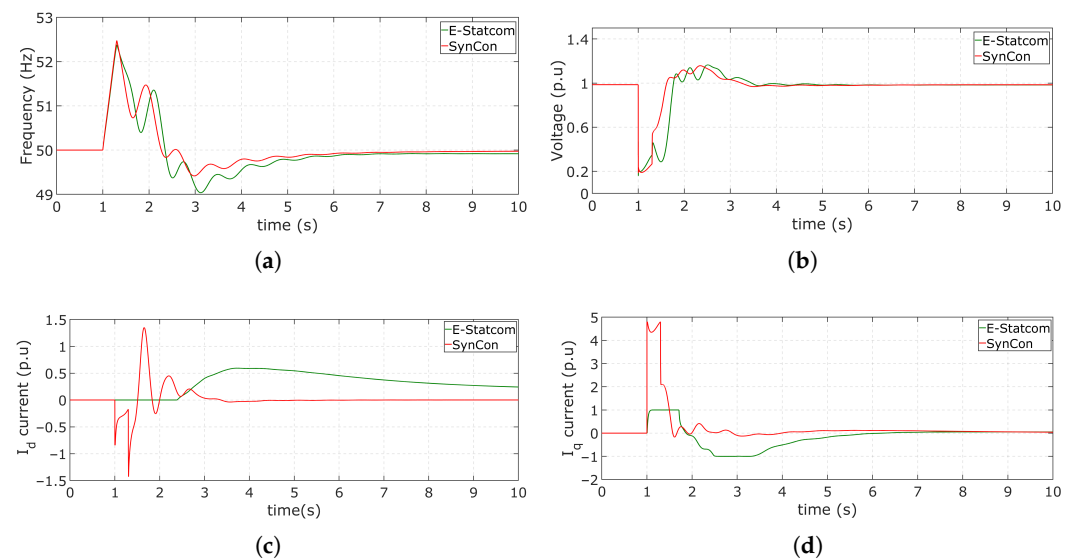


Figure 14. Response to a direct three-phase short circuit at La Oliva bus. (a) Frequency at Punta Grande bus. (b) Voltage at Mácher bus. (c) d-axis current: synchronous condenser and E-STATCOM. (d) q-axis current: synchronous condenser and E-STATCOM.

Figure 14a shows the frequency measured at the Punta Grande bus for the two studied scenarios. It can be observed that in the initial moments during the short circuit, the frequency response exhibits almost identical behavior in both cases. However, once the fault is cleared, the dynamic response when connecting the SynCon stabilizes the system frequency earlier, not reaching such low-frequency oscillations.

Figure 14b presents the voltage magnitude at the Mácher bus, where the compensation devices are connected. The voltage recovery is better with the SynCon because it provides short-circuit current and does not have the current limitations of the E-STATCOM. Figure 14c and Figure 14d show the current components delivered by the SynCon and the E-STATCOM, respectively. Notice in Figure 14d how the quadrature component of the current from the E-STATCOM is constrained by the control system to prevent damage to the power converter, while the current from the SynCon reaches a value close to 5 p.u. Even though the E-STATCOM provides an initial boost due to its rapid response capabilities, its effectiveness is curtailed by built-in current limitations designed to protect its power electronics. This results in a less robust recovery compared to the SynCon. This protective regulation is critical for ensuring the integrity and reliability of the E-STATCOM, but it imposes limitations on its immediate power delivery capabilities during short circuits.

The performance of the E-STATCOM during short-circuit events is inherently limited by its power electronic converters. Unlike the SynCon, which can deliver high short-circuit currents due to its electromechanical nature, the E-STATCOM is constrained by its current-limiting control mechanisms, designed to protect the converters. Figure 14d shows how the quadrature current component of the E-STATCOM is regulated to prevent excessive current flow, whereas the SynCon can inject up to 5 p.u. of current. This protective mechanism ensures the reliability of the E-STATCOM but simultaneously reduces its effectiveness in providing fault ride-through capability. Consequently, the system's voltage recovery is more robust with the SynCon, as shown in Figure 14b, since it can sustain high fault currents without restrictions. While the E-STATCOM offers advantages in frequency stabilization

and fast dynamic response, its ability to support the grid during severe fault conditions is limited compared to the SynCon, requiring supplementary grid-support measures.

In summary, while both the SynCon and E-STATCOM play pivotal roles in frequency and voltage stabilization, their operational characteristics and limitations must be carefully considered when designing power system solutions. The SynCon's ability to provide short-circuit current allows for robust support during direct short circuits, whereas the E-STATCOM's rapid response and energy storage capacity provide quick frequency stabilization, albeit with constraints that must be managed to protect its internal components.

6. Conclusions

The increasing penetration of renewable energy sources in modern power systems, particularly in weak grids, presents significant stability challenges. This study introduces an RMS-based phasorial model of an E-STATCOM with grid-forming (GFM) control, providing a computationally efficient and accurate tool for power system stability analysis. Unlike conventional approaches, the proposed model dynamically adjusts DC voltage in response to active power exchange, capturing energy storage dynamics with greater precision. Additionally, the integration of a power system stabilizer (PSS) enhances frequency stability by mitigating low-frequency oscillations, while the active current limiter (ACL) ensures robust synchronization under large disturbances. Furthermore, the nonlinear modulation function in the exciter improves voltage regulation without introducing harmonic concerns in the RMS framework.

The simulation results on the Lanzarote–Fuerteventura power system confirm that the proposed RMS-based E-STATCOM model significantly enhances frequency stability, voltage control, and fault response. Specifically, the E-STATCOM improves the frequency nadir from 49.35 Hz to 49.55 Hz, compared to the base case, and reduces the settling time by 3.2 s. However, despite these advantages, the E-STATCOM has limitations compared to the synchronous condenser (SynCon). The E-STATCOM delivers approximately 230 MJ to stabilize the grid, while the SynCon supplies around 185 MJ to achieve a similar performance. This energy exchange occurs in both transient and steady-state conditions, emphasizing the longer energy injection period of the E-STATCOM, as it keeps the DC-link voltage relatively constant for an extended duration. Additionally, during short-circuit faults, the E-STATCOM's current injection is limited to 1.2 p.u., whereas the SynCon can provide up to 5 p.u., demonstrating a significantly superior fault ride-through capability.

The findings demonstrate that E-STATCOM with GFM can effectively replace synchronous condensers in scenarios where fast dynamic response and frequency support are required. However, for applications demanding strong fault current injection, SynCons remain a superior choice. As power grids continue their transition toward inverter-dominated architectures, the proposed RMS modeling framework serves as a valuable tool for evaluating and optimizing grid resilience, ensuring stable and reliable system operation in the evolving energy landscape.

Future research will focus on experimental validation and the extension of the model to hybrid energy storage applications, further enhancing its practical applicability.

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